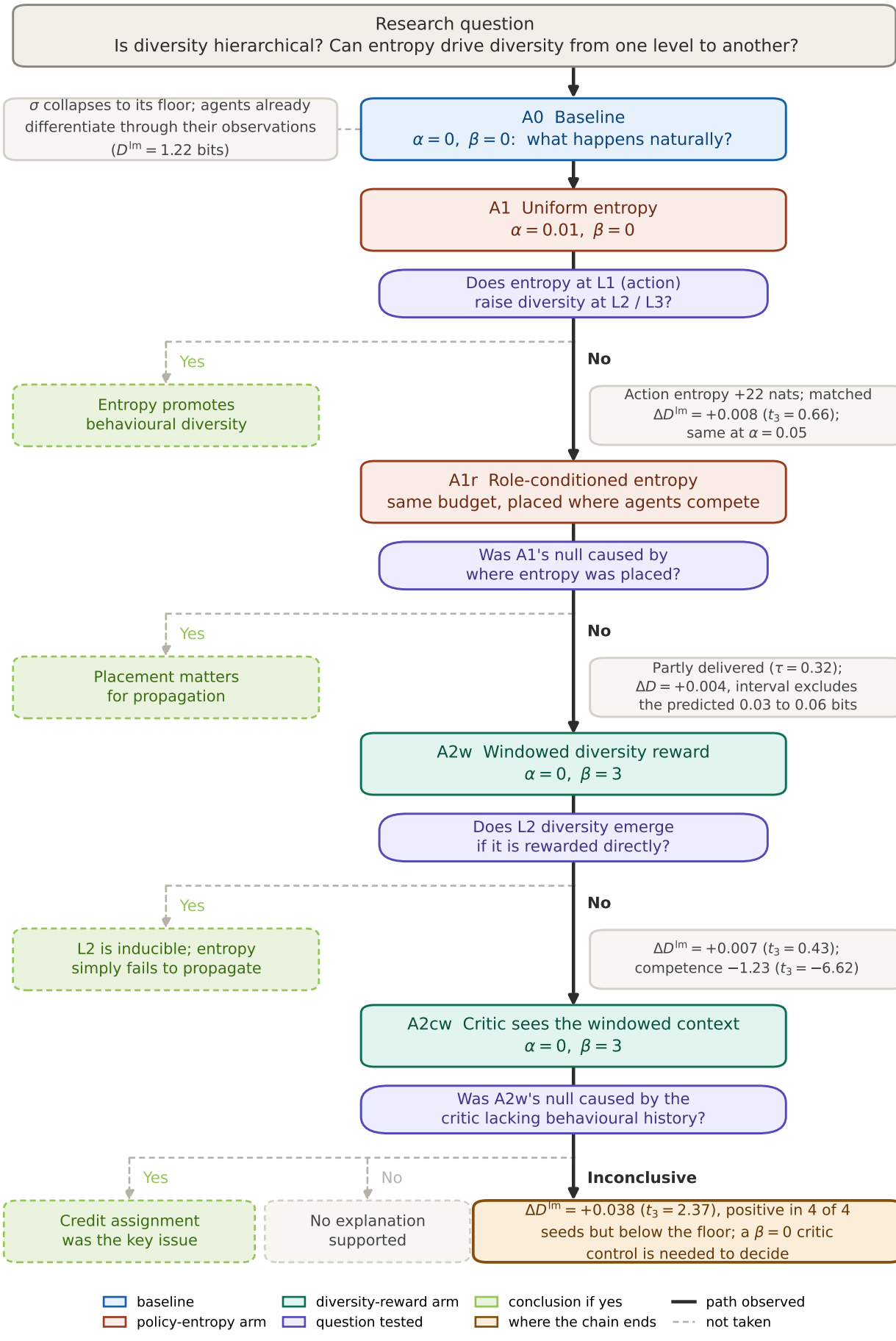
Research question
Is diversity hierarchical? Can entropy drive diversity from one level to another?
σ collapses to its floor; agents already differentiate through their observations ($D^{lm} = 1.22$ bits)
A0 Baseline
$\alpha = 0,\ \beta = 0$: what happens naturally?
A1 Uniform entropy
$\alpha = 0.01,\ \beta = 0$
Does entropy at L1 (action) raise diversity at L2 / L3?
Yes
Entropy promotes behavioural diversity
No
Action entropy +22 nats; matched $\Delta D^{lm} = +0.008$ ($t_3 = 0.66$); same at $\alpha = 0.05$
A1r Role-conditioned entropy
same budget, placed where agents compete
Was A1's null caused by where entropy was placed?
Yes
Placement matters for propagation
No
Partly delivered ($\tau = 0.32$); $\Delta D = +0.004$, interval excludes the predicted 0.03 to 0.06 bits
A2w Windowed diversity reward
$\alpha = 0,\ \beta = 3$
Does L2 diversity emerge if it is rewarded directly?
Yes
L2 is inducible; entropy simply fails to propagate
No
$\Delta D^{lm} = +0.007$ ($t_3 = 0.43$); competence -1.23 ($t_3 = -6.62$)
A2cw Critic sees the windowed context
$\alpha = 0,\ \beta = 3$
Was A2w's null caused by the critic lacking behavioural history?
Yes
Credit assignment was the key issue
No
No explanation supported
Inconclusive
$\Delta D^{lm} = +0.038$ ($t_3 = 2.37$), positive in 4 of 4 seeds but below the floor; a $\beta = 0$ critic control is needed to decide
baseline
policy-entropy arm
diversity-reward arm
question tested
conclusion if yes
where the chain ends
path observed
not taken